

Calculating PUF-PAS Sampling Rates and Effective Volumes in RStudio

A Step-by-Step Workflow

PUF-PAS Effective Volume Model

This R project calculates congener-specific sampling rates and effective sampling volumes for all 209 PCB congeners.

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Original Article

Web-Based Data Analytics Framework for Polyurethane Foam Passive Air Sampling Rate and Effective Volume

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Abstract

Polyurethane foam equipped passive air samplers are a valuable tool for monitoring atmospheric semivolatile organic pollutants, but are often complicated due to the need for an accurate sampling rate (R_s). Here, we describe the development and implementation of a web-based R_s interface that allows for a user-friendly way of determining and visualizing accurate modeled R_s values. The web interface provides access to the previously published model without the need for proprietary software (MATLAB) or coding experience. The precalculated R_s avoids unnecessary computation time, providing the user with fast results and visualizations of the sampler behavior during deployment. Supported by the National Aeronautics and Space Administration's Modern Era Retrospective-Analysis for Research and Applications meteorological data at both 2 and 10 m heights above ground level, users can determine R_s for any location on the globe. Users can select prepopulated compounds (e.g., polychlorinated biphenyls) or manually enter compounds, enter sample information individually or upload them batch-wise, run the model, and export data to a CSV to further integrate with the data analysis workflow. Additionally, an implementation of the same model is available in R, along with scripts to process meteorological data.

Keywords

data analytics, effective sampling rate, effective sampling volume, modeling, PUF-PAS, web interface

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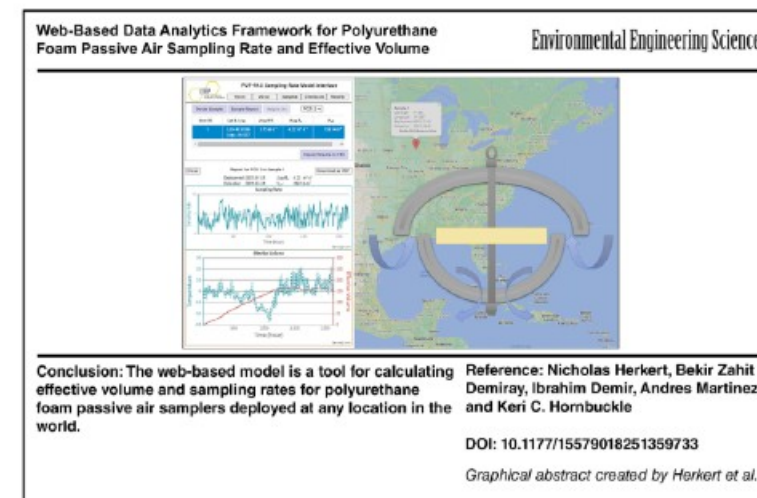
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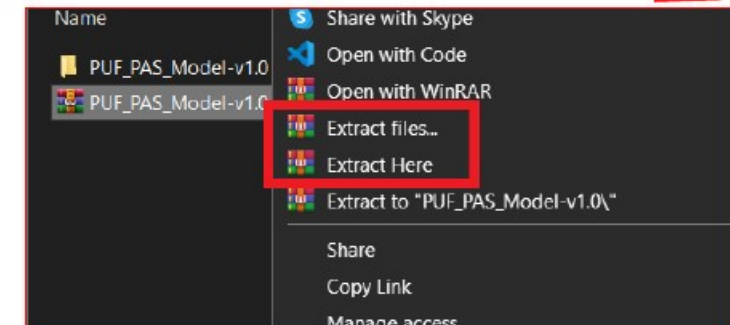
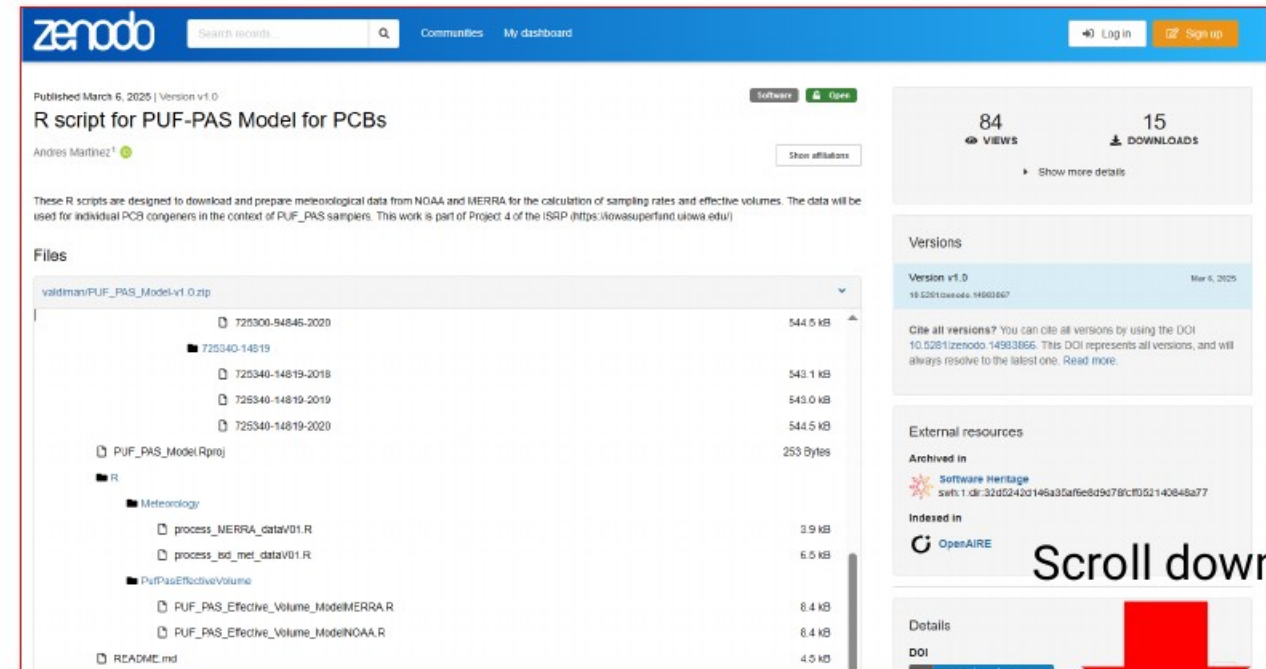


R-project Contents

Item	Function
PUF_PAS_Model.Rproj	Opens the project with the correct working directory
Subfolders.R	Creates the required folder structure
R/Meteorology	Contains NOAA and MERRA meteorological scripts
R/PufPasEffectiveVolume	Contains NOAA and MERRA effective-volume models
Data	Contains PCB physicochemical-property tables
Output	Stores meteorological data and model results
README.md	Basic project instructions

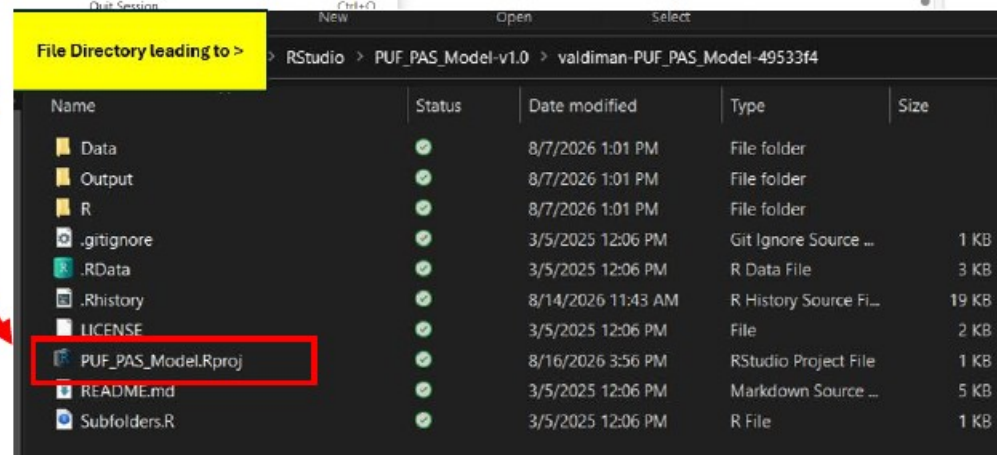
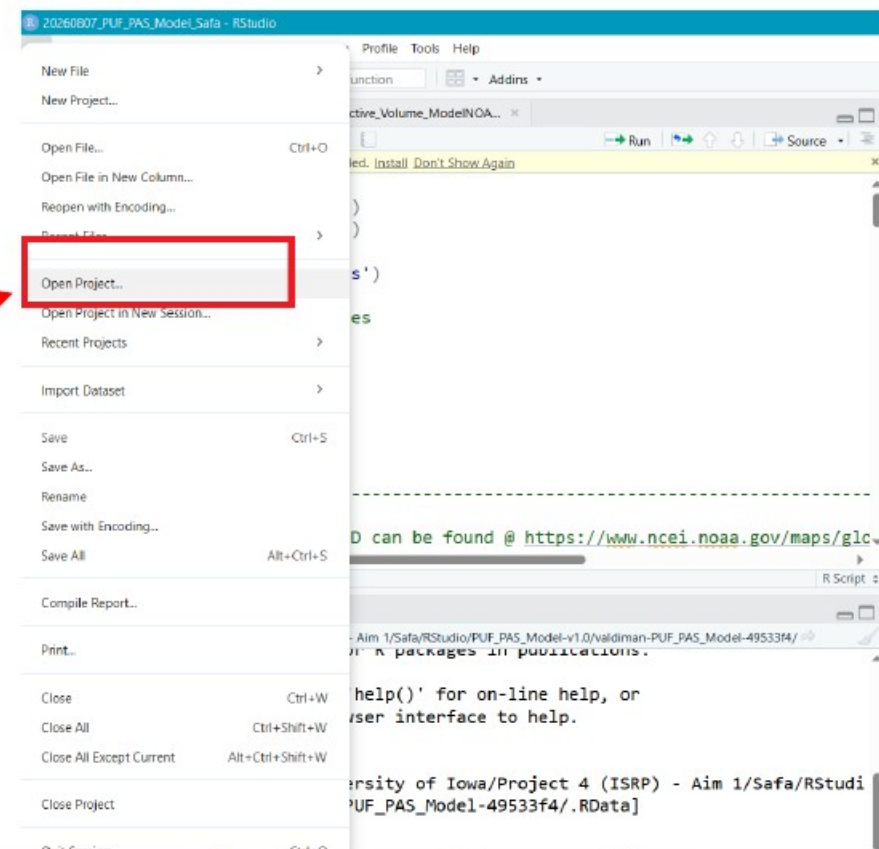
Download and Extract the Project

- **Prerequisite:** Make sure R and RStudio Desktop are installed on your computer before starting
- Go to the **Zenodo repository:**
<https://zenodo.org/records/14983867>
- Download the complete ZIP file to your computer
- Locate the downloaded ZIP file, right-click it, and select Extract All...
- Choose a location for the extracted folder and complete the extraction. You may rename the extracted folder for convenience.
- **Important:** always work from the extracted folder. Do not open or run the R scripts directly from inside the ZIP archive.
- Open RStudio Desktop to begin working with the model files



Open the Project in Rstudio

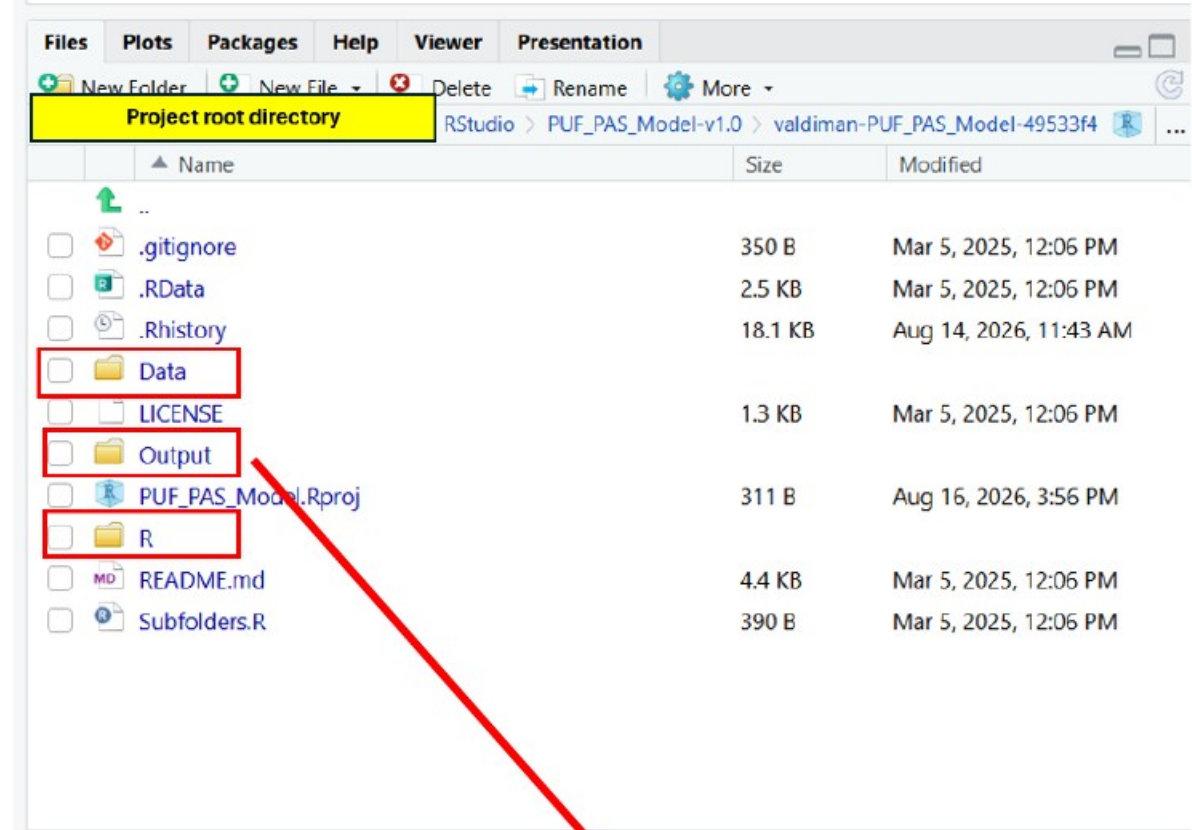
- Open RStudio Desktop
- **Select File → Open Project...**
- Navigate to the extracted project folder
- Select and open **PUF_PAS_Model.Rproj**
- The project will open in RStudio, with the project folder set as the working directory



Confirm that the Project Opened Correctly

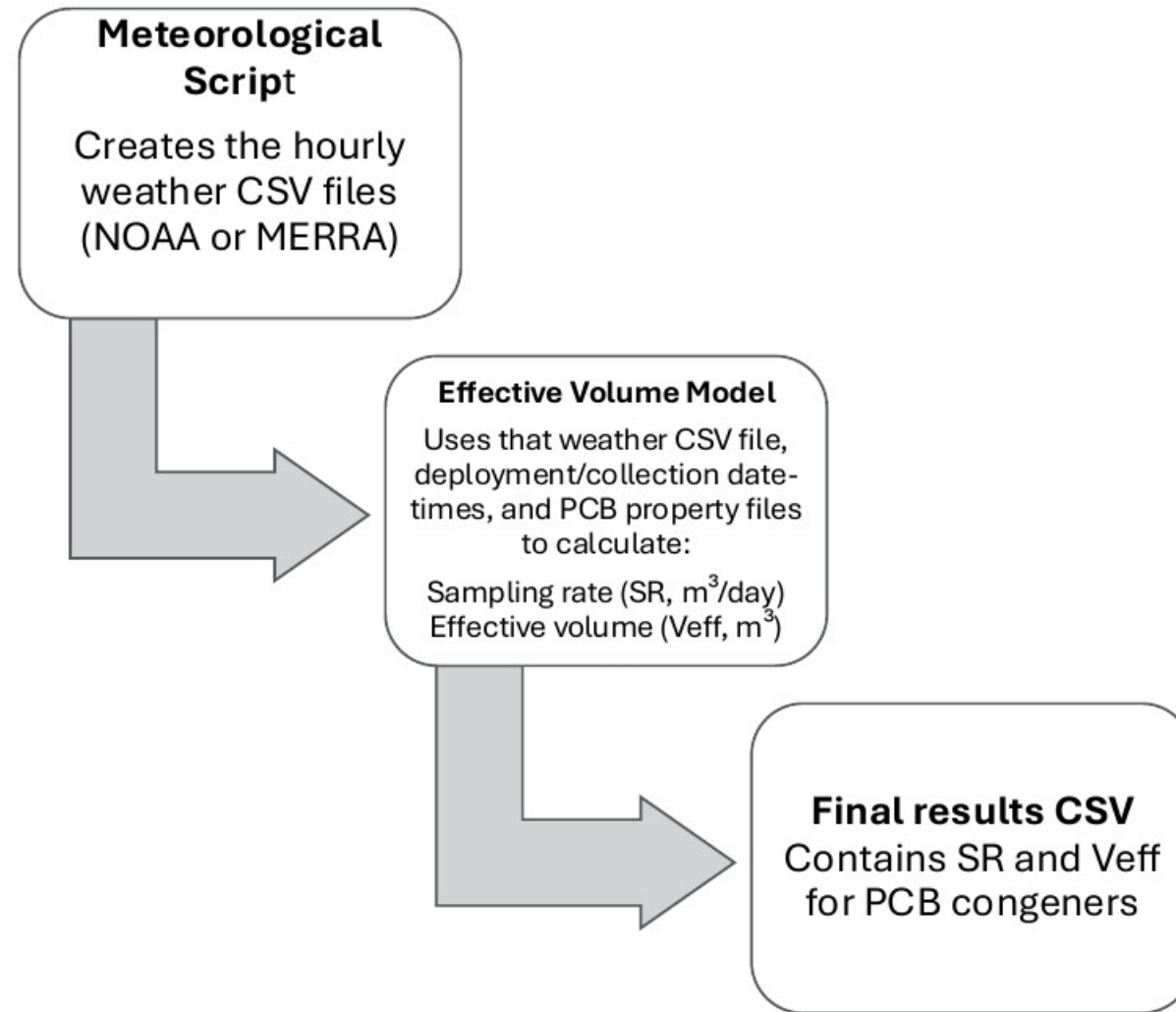
- The **Files** pane should display the project root containing “Data”, “Output”, and “R”

Item	Purpose
Data	PCB physicochemical-property input files
Output	Meteorological files and final model results
R	Meteorological-processing and effective-volume scripts
README.md	Project instructions
Subfolders.R	Creates the required folders if they are missing



Note: The CSV files already stored in **Output** are examples. Delete these example CSVs before running the scripts, but keep the Output folder and all its subfolders

Workflow



Choose GHCNh or MERRA and gather the required inputs

Use NOAA GHCNh when:

A nearby representative GHCNh station is available

- Its record covers the sampling period you need

Use MERRA when:

No suitable nearby NOAA station is available

The required period is unavailable from the station

- MERRA inputs: longitude/latitude and start/end dates

GHCNh required information

Sampling location
GHCNh station ID
Start year and end year
For an incomplete year: end month and end day
Deployment and collection date/time
Time zone

GHCNh is NOAA NCEI's next-generation hourly/synoptic dataset and replaces ISD.

Use the GHCNh product page or NCEI search tools to identify a nearby station and review its period of record.

GHCNh product page

<https://www.ncei.noaa.gov/access/search/datasets/global-historical-climatology-network-hourly/>

Search for the sampling city or location

Select a nearby representative station

- Confirm that the required sampling period is available

Open the NOAA GHCNh or MERRA Meteorological Script

Open the meteorological data processing script

In RStudio, go to the Files pane

Open the R folder, then the Meteorology folder

- Select the script for the data source

NOAA GHCNh

`process_GHCNh_dataV01.R`

Use this script for GHCNh station data.

MERRA

`process_MERRA_dataV01.R`

Use this script when MERRA is the meteorological source.

Double-click the appropriate script to open it in the RStudio Source pane.

For NOAA, the GHCNh script supports two period modes: full calendar years or a user-selected end date for an incomplete/current year.

NOAA GHCNh Workflow: Enter the Station ID, Years and Period Mode

Open “process_GHCNh_dataV01.R”. Install/load the required packages and libraries if needed.

At the user-settings section, edit the station and time-period variables:

```
metdataID <- "USW00014819"

period_mode <- "full_year"
start_year <- 2024
end_year <- 2025
```

Example: USW00014819 is the GHCNh station ID used here for Chicago Midway.

```
period_mode <- "end_date"
start_year <- 2024
end_year <- 2026
end_month <- 9
end_day <- 19
```

Use “full_year” for completed historical years.

Use “end_date” for an incomplete/current year and specify the final month and day needed.

Run the Meteorological Script (NOAA GHCNh)

To run the code:

If packages and libraries were already installed and loaded, no need to run them again.

Save the edited script.

- Click Source to run the entire script.

Tip: Source runs the full script at once; you do not need to paste individual code blocks into the Console.

Full-year mode

Downloads the selected completed calendar years and fills missing hourly observations within the processed period.

End-date mode

Stops at the requested date, but never beyond the last actual GHCNh observation. Missing gaps within the available period can still be filled.

After the script finishes, confirm the station, period, data range, and final saved-file message.

Locate the Processed Meteorological File (NOAA GHCNh)

In RStudio's Files pane, navigate to Output → Data → GHCNh → [station ID].

Locate the processed meteorological CSV file.

The file contains the hourly weather data used by the effective-volume model.

- Note the file path; it will be entered later as `met_data_path`.

Example folder

Output/Data/GHCNh/USW00014819

Typical filename:

```
USW00014819-2024-2025-GHCNh-filled.csv
```

StationID-StartYear-EndYear-GHCNh-filled.csv

For an incomplete-year run, the filename still uses the selected start and end years; the actual data range is controlled by `period_mode` and the available GHCNh observations.

MERRA Workflow: Enter the Coordinates and Dates

- Open: “**process_MERRA_dataV01.R**”
- Install the required R packages if they are not already installed (lines 2 – 4)
- Load the required libraries before running the data-processing code (lines 8 – 10)
- Press **Ctrl + F** and search for “**lonlat**” (line 15)
- Replace the example coordinates with the coordinates of your sampling location. Enter the coordinates in longitude, latitude order. The coordinates can be obtained using **Google Earth**
- Replace the sampling years (yyyy-mm-dd)
- Search for **output_dir** (line 102) and replace the example site name in the output folder, and search for the **output_file_path** (line 108) and replace CSV filename (example: OHareAirport-2018-2020.csv)

Variable	Replace with
lonlat	Sampling-site longitude and latitude
start_date	First meteorological date required
end_date	Last meteorological date required
output_dir	Site-specific output folder name
output_file_path	Site- and period-specific CSV filename

```
5  
6 # Load necessary libraries  
7 {  
8   library(dplyr)  
9   library(nasapower)  
10  library(zoo)  
11 }  
12  
13 # Location and time selection  
14 # Define coordinates. E.g., O'Hare International Airport, Chicago (Longitude, Latitude)  
15 lonlat <- c(-87.904722, 41.978611)  
16  
17 # Define the start and end date  
18 start_date <- "2018-01-01"  
19 end_date <- "2020-12-31"  
20
```

```
100  
101 # Create output directory to store data  
102 output_dir <- file.path("Output/Data/MERRA", "OHareAirport") # Need to adjust name, e.g., O'Hare Airport  
103 if (!dir.exists(output_dir)) {  
104   dir.create(output_dir, recursive = TRUE)  
105 }  
106  
107 # Save the final filled data without 'T2MDEW'  
108 output_file_path <- file.path(output_dir, "OHareAirport-2018-2020.csv") # Need to adjust name  
109 tryCatch({  
110   write.csv(filled_data, file = output_file_path, row.names = FALSE)  
111   cat(sprintf("Final filled data saved to %s\n", output_file_path))  
112 }, error = function(e) {  
113   cat("Error saving the file:", e$message, "\n")  
114 })
```

Run the MERRA Meteorological Script

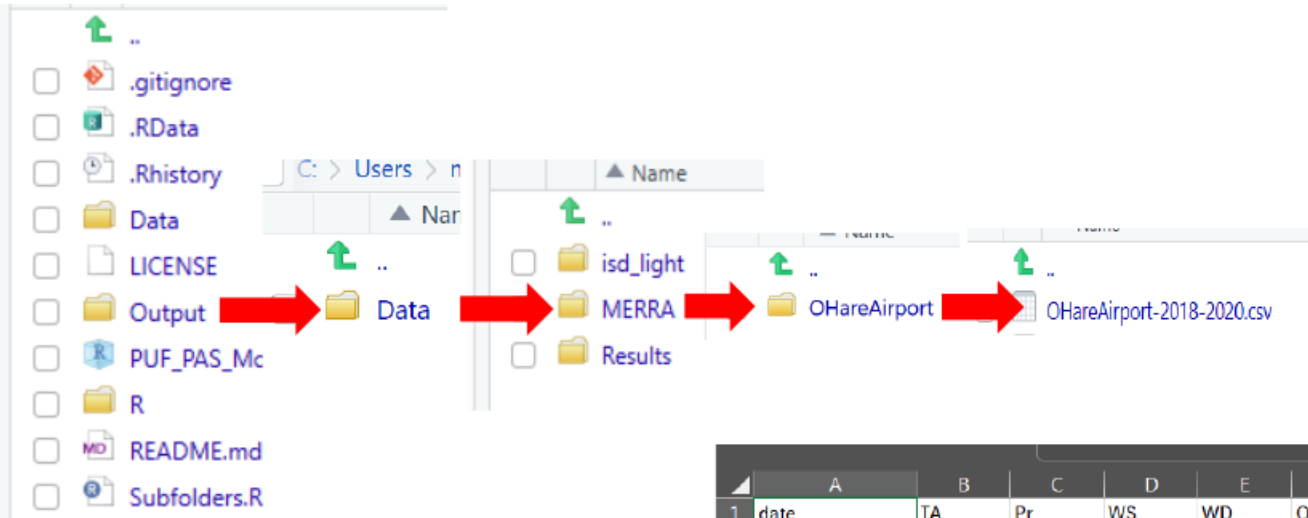
- To run the code:
- If packages and libraries were already installed and loaded, no need to run it again (see instruction on previous slide)
- Confirm the selected dates and coordinates in the **POWER** summary
- For this O'Hare 2018–2020 example, the processed dataset contains 26,304 hourly observations and six required variables
- Confirm completion using the **final saved-file** message

```
process_isd_met_dataV01.R x process_MERRA_dataV01.R x PUF_PAS_Effective_Volume_ModelNOA... x
Source on Save
Packages nasapower and zoo required but are not installed. Install Don't Show Again
94:15 Download data from MERRA-2
R Script

Console Terminal Background Jobs
R - R 4.5.1 - G:/University of Iowa/Project 4 (ISRP) - Aim 1/Safa/RSstudio/PUF_PAS_Model-v1.0/valdiman-PUF_PAS_Model-4953314/

POWER.In [26,304 x 11] (S3: POWER.Info/tbl_df/tbl/data.frame)
$ LON : num [1:26304] -87.9 -87.9 -87.9 -87.9 -87.9 ...
$ LAT : num [1:26304] 42 42 42 42 42 ...
$ YEAR : num [1:26304] 2018 2018 2018 2018 2018 ...
$ MO : num [1:26304] 1 1 1 1 1 1 1 1 1 ...
$ DY : num [1:26304] 1 1 1 1 1 1 1 1 1 ...
$ HR : num [1:26304] 0 1 2 3 4 5 6 7 8 9 ...
$ T2M : num [1:26304] -16.7 -17.8 -18.7 -19.7 -20.5 ...
$ T2MDEW : num [1:26304] -19 -19.9 -20.7 -21.4 -22.1 ...
$ PS : num [1:26304] 101 101 101 101 101 ...
$ WD10M : num [1:26304] 316 314 312 311 310 ...
$ WS10M : num [1:26304] 5.97 6.06 6.18 6.3 6.26 6.3 6.38 6.25 6.11 6.15 ...
- attr(*, "spec")=
.. cols(
.. YEAR = col_double(),
.. MO = col_double(),
.. DY = col_double(),
.. HR = col_double(),
.. T2M = col_double(),
.. T2MDEW = col_double(),
.. PS = col_double(),
.. WD10M = col_double(),
.. WS10M = col_double()
.. )
- attr(*, "problems")=<externalptr>
- attr(*, "POWER.Info")= chr "NASA/POWER Source Native Resolution Hourly Data "
- attr(*, "POWER.Dates")= chr "Dates (month/day/year): 01/01/2018 through 12/31/2020 in UTC"
- attr(*, "POWER.Location")= chr "Location: Latitude 41.9786 Longitude -87.9047 "
- attr(*, "POWER.Elevation")= chr "Elevation from MERRA-2: Average for 0.5 x 0.625 degree lat/lon region = 231.39 meter s"
- attr(*, "POWER.Climate_zone")= chr "The value for missing source data that cannot be computed or is outside of the sou rces availability range: NA "
- attr(*, "POWER.Missing_value")= chr "Parameter(s): "
- attr(*, "POWER.Parameters")= chr "T2M MERRA-2 Temperature at 2 Meters (C) ;\n T2MDEW MERRA-2 Dew/Frost Poin t at 2 Meters (C) ;\n PS |__truncated__ NULL
'data.frame': 26304 obs. of 6 variables:
$ date: POSIXct, format: "2018-01-01 00:00:00" "2018-01-01 01:00:00" ...
$ TA : num 256 255 254 253 253 ...
$ Pr : num 100740 100760 100810 100840 100880 ...
$ WS : num 5.97 6.06 6.18 6.3 6.26 6.3 6.38 6.25 6.11 6.15 ...
$ WD : num 316 314 312 311 310 ...
$ QV : num 1.55e-04 1.28e-04 1.08e-04 9.07e-05 7.76e-05 ...
Final filled data saved to Output/Data/MERRA/OHareAirport/OHareAirport-2018-2020.csv
```

Locate the Processed Meteorological file (MERRA)



- In RStudio's Files pane, navigate to (MERRA example): **Output** → **Data** → **MERRA** → **OHareAirport**
- Locate the processed meteorological data file. In our example:
OHareAirport-2018-2020.csv
- This CSV contains the processed hourly MERRA-2 meteorological data for the O'Hare 2018–2020 example
- Note the location where the file was saved, as its file path will be entered later as “**met_data_path**” in the MERRA effective-volume model
- MERRA filenames generally follow: SamplingLocation-StartYear-EndYear.csv; For example, “OHareAirport-2018-2020.csv”

Processed
MERRA-2
CSV
opened in
Excel



	A	B	C	D	E	F
1	date	TA	Pr	WS	WD	QV
2	1/1/2018	256.45	100740	5.97	315.8	0.000155
3	1/1/2018 1:00	255.39	100780	6.06	313.7	0.000128
4	1/1/2018 2:00	254.44	100810	6.18	312	0.000108
5	1/1/2018 3:00	253.49	100840	6.3	310.8	9.07E-05
6	1/1/2018 4:00	252.65	100880	6.26	309.8	7.76E-05
7	1/1/2018 5:00	251.98	100910	6.3	310.7	6.77E-05
8	1/1/2018 6:00	251.54	100930	6.38	311.3	6.18E-05
9	1/1/2018 7:00	251.13	100970	6.25	310.7	5.75E-05
10	1/1/2018 8:00	250.76	100990	6.11	308	5.43E-05
11	1/1/2018 9:00	250.28	100990	6.15	305	5.05E-05
12	1/1/2018 10:00	249.85	100990	6.25	302.1	4.74E-05
13	1/1/2018 11:00	249.55	101020	6.34	299.4	4.54E-05
14	1/1/2018 12:00	249.51	101060	6.5	299.2	4.51E-05
15	1/1/2018 13:00	249.92	101110	6.7	301.7	4.79E-05
16	1/1/2018 14:00	251.3	101150	7.25	305.5	5.97E-05
17	1/1/2018 15:00	252.58	101190	7.18	307.8	7.30E-05
18	1/1/2018 16:00	253.97	101190	6.82	307.8	9.14E-05
19	1/1/2018 17:00	255.67	101140	6.47	307.1	0.00012
20	1/1/2018 18:00	257	101070	6.15	303.9	0.000151
21	1/1/2018 19:00	257.88	101040	5.98	301.3	0.000176
22	1/1/2018 20:00	258.22	101050	5.83	299.6	0.000188
23	1/1/2018 21:00	257.62	101070	5.44	297.2	0.000176
24	1/1/2018 22:00	255.52	101090	4.74	292.3	0.000132
25	1/1/2018 23:00	254.1	101120	4.57	285.5	0.00011
26	1/2/2018	253.25	101130	4.58	280.6	9.90E-05

Open the Effective Volume Model Script (NOAA GHCNh or MERRA)

In RStudio, go to the Files pane → R → PufPasEffectiveVolume.

Select the appropriate model script:

NOAA GHCNh

PUF_PAS_Effective_Volume_ModelGHCNh.R

MERRA

PUF_PAS_Effective_Volume_ModelMERRA.R

Double-click the appropriate script to open it in the RStudio Source pane.

The effective-volume model uses the processed meteorological CSV, deployment/collection date-times, and the PCB property files to calculate congener-specific sampling rates and effective volumes.

NOAA GHCNh Effective Volume Model Script (PUF_PAS_Effective_Volume_ModelGHCNh.R)

To run the code

No additional packages are required for the effective-volume model.

Run the function definitions once after starting a new R session.

- Go to the “Run the Model” section and enter the deployment dates.

```
start_date = "2026-03-01 01:00:00"  
end_date   = "2026-04-10 01:00:00"  
  
met_data_path =  
"Output/Data/GHCNh/USW00014819/USW00014819-  
2026-2026-GHCNh-filled.csv"
```

Check that `met_data_path` points to the processed GHCNh CSV, `pcb_properties_path` points to the PCB properties file, and `file_path` points to the desired results folder.

```
file_path =  
"Output/Data/Results/GHCNh/  
VefChicagoMidwayAirport2026.csv"
```

MERRA Effective Volume Model Script (PUF_PAS_Effective_Volume_ModelMERRA.R)

To run the code

- No additional R packages or libraries are required
- Run the function definitions (approximately lines 6–242)
 - These functions contain the calculations needed for the effective-volume model
 - You do not need to run each individual function separately; you can select and run the entire section at once
- Go to line 244, “**Run the Model.**” In this section:
 1. Enter the start and end dates of your deployment using the format **yyyy-mm-dd hh:mm:ss**. Enter 00 for minutes and seconds when they are not known or not needed. For example: 2018-12-01 01:00:00
 2. Check that “**met_data_path**” points to the correct location of your processed meteorological data (line 249)
 3. Check that **pcb_properties_path** points to the correct PCB properties file. We recommend using “**Data/PCB_Properties_MW_DU_KOA.csv**”
 4. Check that **file_path** specifies the correct location and filename for the model output, under “**Output/Data/Results/MERRA/....**”

```
235 colnames(outData) <- column_names
236
237 # Create the directory if it does not exist
238 dir.create(dirname(file_path), recursive = TRUE)
239
240 # Save the data frame to a CSV file
241 write.csv(outData, file = file_path, row.names = FALSE)
242 }
243
244 # Run the Model -----
245 PUF_PAS_Effective_Volume_Calculations(
246   start_date = "2018-12-01 01:00:00",
247   end_date = "2019-01-10 01:00:00",
248   # Need to select the folder and name of the file where the met data is stored
249   met_data_path = "Output/Data/MERRA/OHareAirport/OHareAirport-2018-2020.csv",
250   pcb_properties_path = "Data/PCB_Properties_MW_DU_KOA.csv",
251   pcb_ppLFR_path = "Data/PCB_LFR_descriptors.csv",
252   # Need to add the folder where the result is going to be saved
253   # It should be metdataID-year
254   file_path = "Output/Data/Results/MERRA/VefOHareAirport2018-2020.csv"
255 )
256
```

Final Model Output

Locate the effective-volume model results in Output → Data → Results → GHCNh.

Open the CSV file generated by the effective-volume model.

- You can open the file in RStudio or Microsoft Excel to review the results.

NOAA GHCNh results folder

Output/Data/Results/GHCNh

The output file contains 2 data rows (Veff and SR) plus a header row, with the PCB congeners represented across the columns.

```
VefChicagoMidwayAirport2026.csv
```

Use a unique, descriptive filename that identifies the sampling location and deployment period.

Rerun the Model for Another Location and/or Deployment Period

To run the effective-volume model again:

Process the meteorological data first using the appropriate NOAA GHCNh or MERRA script.

In the Run the Model section, update start_date and end_date.

Update met_data_path to the processed meteorological CSV.

Update file_path to a new, descriptive output filename.

The PCB property files generally do not need to be changed.

- Run the “Run the Model” section again.

GHCNh current-year option

When the year is incomplete, use period_mode = "end_date" in the GHCNh meteorological script and specify end_month and end_day.

Unique outputs

Use a distinct filename for each location and deployment period so previous results are not overwritten.

If the function definitions are still loaded in the current R session, they do not need to be rerun. After restarting RStudio, run the function definitions before the model.